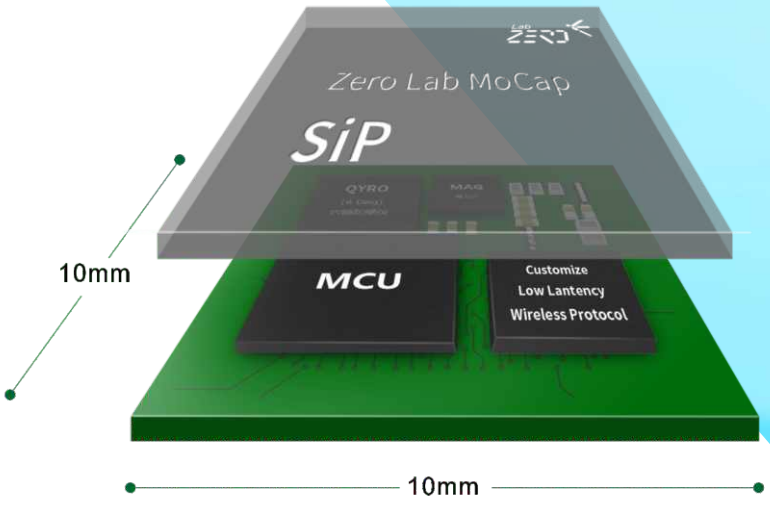
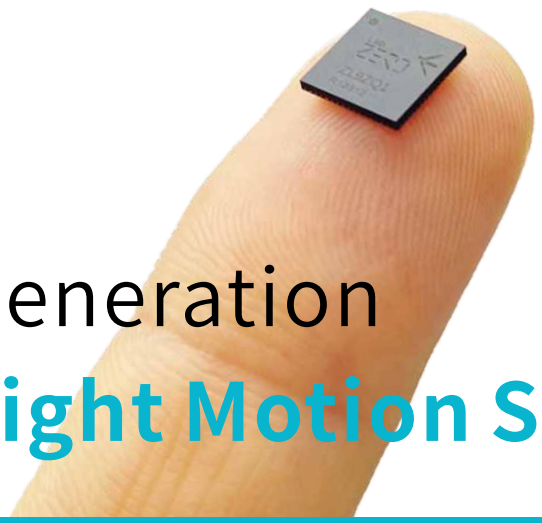
Lab ZERO

We Make Reality Better



Ultralight Motion Sensor



ZL9NSQ

New Generation
Ultralight Motion Sensor Chip

Zero Lab Technologies.,Ltd

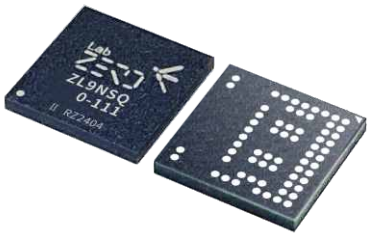
Zero Lab Committed to building Telepresence technology solutions,
through motion capture technology, digitizing humans Auditory sense, Visual perception,
Motion and haptic etc, integrating AI naturally to expand human-machine interaction capability.

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Powered by

ZL9NSQ



Technical Specification

Highlights

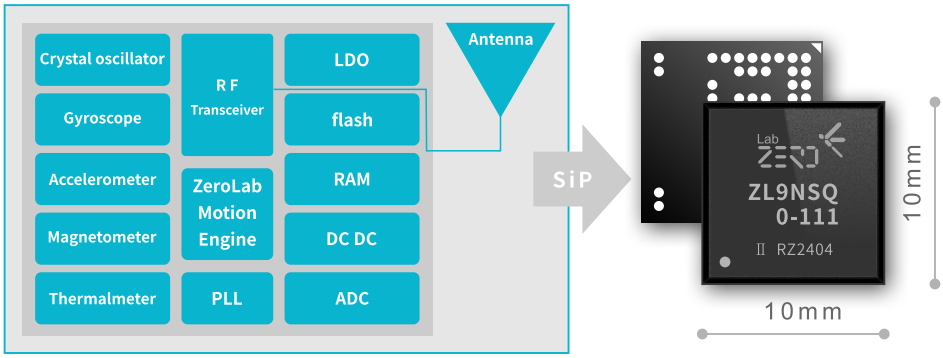
	ZL9NSQ Motion Sensor Chip	Traditional Solutions
Size	10 × 10 mm All-in-one SiP	40 × 40 mm Multiple Chips module
Power consumption	50 mW	200 mW
Cost	$	$$$$$$$$$$

Features

- **The industry's first ultra-small 10 x 10 x 2mm motion capture SiP chip**
- **Adopt advanced Chiplet integrated circuit technology with built-in high performance antenna**
- **Built-in RF 2.4 GHz SoC, supports Bluetooth BLE 5.2 and Zerolab custom low latency communication protocol**
- **Built-in 3-axis accelerometer, 3-axis gyroscope, 3-axis magnetometer**
- **3.3V single power supply (no external circuit required), lower voltage supply, longer standby time**

The Zero Lab ultra-thin motion capture sensor chip ZL9NSQ is designed for low-power, high-precision all-in-one chip-level system solutions in the field of motion capture. Using the industry's latest Chiplet (heterogeneous packaging integrated circuit) technology, massive integration has been achieved in an ultra-small 10 x 10 x 2mm package,No need any external circuit except power supply, intergrated advacne sensor fusion algorithm,plug and play, effectively reducing solution cost and speed up R&D process.

The ZL9NSQ ultra-thin motion capture chip can be embedded in smart wearable devices such as sport suit and wristbands, lead the smart wearable application to the new era,encourage the openmind innovation.



Product Description

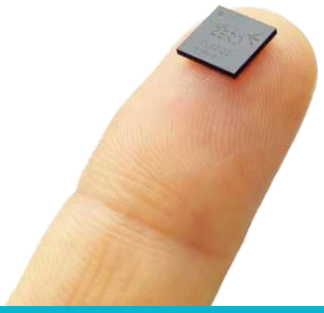
Item	Instructions
Voltage	3.3 V
Electric current	< 15 mA
Built-in attitude sensor	3 axis acceleration, 3 axis angular velocity, 3 axis magnetometer
Acceleration range	± 2 / ± 4 / ± 8 / ± 16 g
Magnetometer range	± 800 uT
gyroscope range	± 256 / 512 / 1024 / 2048 dps
Attitude accuracy	Static accuracy: X / Y axis 0.05°, Z axis 0.5;Dynamic accuracy:X / Y axis 0.1°, Z axis 0.5°
Output data format	Attitude quaternion/attitude Euler Angle/angular velocity/acceleration/linear acceleration/magnetic field/battery level
Output interface	UART / I^2C / SPI / BLE 5.2 / Zerolab Private protocol
Extension interface	Analog signal input (AD0 The default value is battery level check)
External Antenna	Supported
Chip size	10 mm X 10 mm X 2 mm

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Chip Size

ZL9NSQ chip adopted the most advance Chiplet manufacturing process, which significant reduced chip size.Used industry-leading materials and processes to successfully achieved the smallest size in the industry,without performance degradation ZL9NSQ chip are lighter, thinner and smaller,can be perfectly embedded in your miniaturized smart devices.





ZL9NSQ Size



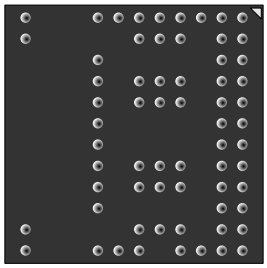
Industry PCB Module Size

Package Type

The ZL9NSQ chip adopted WBBGA (wafer-level ball grid array) packaging technology,great improves the reliability of the chip assemble, enhance the stability for use in moist environment, and guarantee long service cycle.



Top View



Bottom View



Side View

Application Scenarios

The ZL9NSQ chip is an industry-leading development solution for motion capture, analysis, and reporting. Built in BLE and custom wireless communication protocol enable real time communication with computer and hand hold device. ZL9NSQ serve for different application like smart wearable moniotor, virtual reality, remote positioning and rehabilitation etc, special tuning embedded algorithm provide user out of ordinary intelligent experiences.



Motion Data Acquisition
Motion capture, sports, rehabilitation

Virtual reality interaction
Metaverse、XR、MR

Remote location tracking
Logistics, transportation, fire protection

Machine intelligent perception
Robots, drones, the IoT

WeChat Website

Zero Lab Technologies ., Ltd
Add: Room 908, Building C, Tiansheng Kechuang Plaza, No.421 Zhengli Road, Yangpu District, Shanghai
Tel:021-55809628
Email:info@zero-lab.tech